

Modelling and System Analysis

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1 Introduction

This report studies the dynamics of a mechanical system composed of two point masses connected by a linear spring, evolving in a plane in the absence of gravity. The mechanical system considered in this work is illustrated in Figure 1.

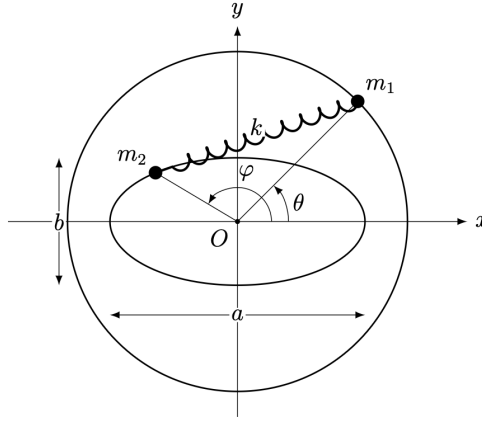


Figure 1: Schematic representation of the mechanical system.

The first mass m_1 is constrained to slide without friction along a circular trajectory of radius R centered at the origin O . The second mass m_2 is constrained to slide on an ellipse, also centered at O , whose major and minor axes are respectively a and b . Unlike m_1 , the mass m_2 is subject to a viscous friction force proportional to its velocity.

The two masses are connected by a linear spring of stiffness k and natural length l_0 , which introduces a coupling between the two trajectories. The system is not subject to gravity, so the only potential energy present in the system is the elastic energy stored in the spring.

The geometric constraints imposed on each mass reduce the system to two degrees of freedom, fully described by the generalized coordinates

$$q_1 = \theta, \quad q_2 = \varphi,$$

where θ denotes the angular position of m_1 on the circle and φ denotes the parametric angle describing the position of m_2 on the ellipse.

This first report successively establishes the kinematic description of the system, derives the Lagrangian, and incorporates the friction force through the generalized force formalism.

2 Kinematic Description

The motion takes place in the plane (x, y) and the origin O is chosen as the reference point for the position vectors.

Mass m_1 . Since m_1 moves on a circle of radius R , its position vector relative to the origin is given by

$$\vec{r}_1 = \begin{pmatrix} R \cos \theta \\ R \sin \theta \end{pmatrix}. \quad (1)$$

Mass m_2 . The mass m_2 moves on an ellipse centered at O with semi-major axis $a/2$ and semi-minor axis $b/2$. A convenient parametric representation of the ellipse leads to the position vector

$$\vec{r}_2 = \begin{pmatrix} \frac{a}{2} \cos \varphi \\ \frac{b}{2} \sin \varphi \end{pmatrix}. \quad (2)$$

Equations (1) and (2) fully describe the spatial configuration of the two masses.

The spring connects the two masses and its deformation depends on the distance between them. The vector connecting m_1 and m_2 is defined as

$$\vec{r}_{12} = \vec{r}_1 - \vec{r}_2. \quad (3)$$

Substituting the expressions of the position vectors given in (1) and (2) into (3) yields

$$\vec{r}_{12} = \begin{pmatrix} R \cos \theta - \frac{a}{2} \cos \varphi \\ R \sin \theta - \frac{b}{2} \sin \varphi \end{pmatrix}. \quad (4)$$

The instantaneous length of the spring corresponds to the Euclidean norm of this vector as

$$l = \|\vec{r}_{12}\| = \sqrt{\left(R \cos \theta - \frac{a}{2} \cos \varphi\right)^2 + \left(R \sin \theta - \frac{b}{2} \sin \varphi\right)^2}. \quad (5)$$

The velocities of the masses are obtained by differentiating the position vectors with respect to time.

Differentiating Equation (1) gives

$$\dot{\vec{r}}_1 = \begin{pmatrix} -R \sin \theta \dot{\theta} \\ R \cos \theta \dot{\theta} \end{pmatrix}. \quad (6)$$

The squared magnitude of this velocity is therefore

$$v_1^2 = R^2 \dot{\theta}^2. \quad (7)$$

Similarly, differentiating Equation (2) leads to

$$\dot{\vec{r}}_2 = \begin{pmatrix} -\frac{a}{2} \sin \varphi \dot{\varphi} \\ \frac{b}{2} \cos \varphi \dot{\varphi} \end{pmatrix}. \quad (8)$$

The squared magnitude of this velocity becomes

$$v_2^2 = \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi\right) \dot{\varphi}^2. \quad (9)$$

3 Lagrangian Formulation

The kinetic energy of the system is the sum of the kinetic energies of the two masses,

$$K = K_1 + K_2 = \frac{1}{2} m_1 R^2 \dot{\theta}^2 + \frac{1}{2} m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi\right) \dot{\varphi}^2. \quad (10)$$

Since the system is not subject to gravity, the only potential energy arises from the deformation of the spring.

The elastic potential energy stored in the spring is

$$P = \frac{1}{2} k (l - l_0)^2. \quad (11)$$

The spring length l appearing in Eq. (11) is given by Eq. (5). This expression introduces a nonlinear coupling between the generalized coordinates θ and φ .

The Lagrangian of the system is defined as the difference between kinetic and potential energies,

$$L = K - P. \quad (12)$$

Substituting the expressions (10) and (11) yields the Lagrangian of the system,

$$L = \frac{1}{2}m_1 R^2 \dot{\theta}^2 + \frac{1}{2}m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi}^2 - \frac{1}{2}k(l - l_0)^2, \quad (13)$$

where l is the instantaneous spring length defined in (5). The nonlinear coupling between θ and φ appears through l .

The equations of motion will then be obtained by applying the Euler–Lagrange equations,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i, \quad i \in \{\theta, \varphi\}. \quad (14)$$

where Q_i represents the generalized forces acting on the system.

In the present system, a viscous friction force acts on the mass m_2 . This friction is assumed to be proportional to the velocity of the mass and opposite to its motion. The friction force is therefore modeled as

$$\vec{F}_f = -\alpha \dot{\vec{r}}_2, \quad (15)$$

where α denotes the viscous damping coefficient and $\dot{\vec{r}}_2$ is the velocity of the mass m_2 , given by equation (8). The generalised force associated with the coordinate φ is obtained via the virtual work principle,

$$Q_\varphi = \vec{F}_f \cdot \frac{\partial \vec{r}_2}{\partial \varphi}. \quad (16)$$

Since $\dot{\vec{r}}_2 = \dot{\varphi} \frac{\partial \vec{r}_2}{\partial \varphi}$, this reduces to

$$Q_\varphi = -\alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi}. \quad (17)$$

This dissipative force introduces an additional term in the Euler–Lagrange equation associated with the coordinate φ . The equations of motion therefore become

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0, \quad (18)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\varphi}} \right) - \frac{\partial L}{\partial \varphi} + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0. \quad (19)$$

Since the friction acts only on the mass m_2 , the coordinate θ is not directly affected by this dissipative force.

4 Equations of Motion

The equations of motion of the system are obtained by applying the Euler–Lagrange equations (14) to the Lagrangian defined in (12).

The Euler–Lagrange equation associated with the coordinate θ gives

$$m_1 R^2 \ddot{\theta} + k(l - l_0) \frac{\partial l}{\partial \theta} = 0. \quad (20)$$

Similarly, applying the Euler–Lagrange equation to the coordinate φ and including the generalised friction force (17) yields

$$\begin{aligned} \frac{d}{dt} \left[m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} \right] - m_2 \frac{a^2 - b^2}{4} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ + k(l - l_0) \frac{\partial l}{\partial \varphi} + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0. \end{aligned} \quad (21)$$

Equations (20) and (21) form a system of two nonlinear coupled differential equations describing the dynamics of the system.

4.1 Derivatives of the spring length

To make the equations of motion explicit, the derivatives of the spring length l with respect to the generalized coordinates must be computed.

From (5), the derivative with respect to θ becomes

$$\frac{\partial l}{\partial \theta} = \frac{-R \sin \theta \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) + R \cos \theta \left(R \sin \theta - \frac{b}{2} \sin \varphi \right)}{l}. \quad (22)$$

Similarly, the derivative with respect to φ is

$$\frac{\partial l}{\partial \varphi} = \frac{\frac{a}{2} \sin \varphi \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) - \frac{b}{2} \cos \varphi \left(R \sin \theta - \frac{b}{2} \sin \varphi \right)}{l}. \quad (23)$$

4.2 Final Equations of Motion

Substituting the derivatives into the Euler-Lagrange equations yields the explicit equations of motion.

$$m_1 R^2 \ddot{\theta} + \frac{k(l-l_0)}{l} \left[-R \sin \theta \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) + R \cos \theta \left(R \sin \theta - \frac{b}{2} \sin \varphi \right) \right] = 0 \quad (24)$$

$$\begin{aligned} m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \ddot{\varphi} + m_2 \frac{a^2 - b^2}{4} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ + \frac{k(l-l_0)}{l} \left[\frac{a}{2} \sin \varphi \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) - \frac{b}{2} \cos \varphi \left(R \sin \theta - \frac{b}{2} \sin \varphi \right) \right] + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0 \end{aligned} \quad (25)$$

Equations (24) and (25) constitute the complete nonlinear equations of motion of the system. These equations describe the coupled dynamics of the two masses connected by the spring, including the viscous friction acting on the second mass.

5 Non-dimensionalization

Before proceeding to non-dimensionalization, the geometric parameters by the problem statement,

$$a = \frac{3}{4} R, \quad b = \frac{1}{4} R, \quad l_0 = R,$$

are substituted into the final equations of motion derived in the previous section. The position vectors become

$$\vec{r}_1 = \begin{pmatrix} R \cos \theta \\ R \sin \theta \end{pmatrix}, \quad (26)$$

$$\vec{r}_2 = \begin{pmatrix} \frac{3R}{8} \cos \varphi \\ \frac{R}{8} \sin \varphi \end{pmatrix}. \quad (27)$$

The instantaneous spring length is given by

$$\begin{aligned} l &= \sqrt{\left(R \cos \theta - \frac{3R}{8} \cos \varphi \right)^2 + \left(R \sin \theta - \frac{R}{8} \sin \varphi \right)^2} \\ &= R \sqrt{1 - \frac{3}{4} \cos \theta \cos \varphi - \frac{1}{4} \sin \theta \sin \varphi + \frac{1}{64} (9 \cos^2 \varphi + \sin^2 \varphi)}. \end{aligned} \quad (28)$$

Substituting the geometric parameters into equation (24) and simplifying, the equation of motion for θ reduces to

$$\boxed{m_1 R^2 \ddot{\theta} + \frac{k R^2}{8} \frac{l - R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) = 0.} \quad (29)$$

Similarly, substituting the geometric parameters into equation (25) and simplifying, the equation of motion for φ reduces to

$$\boxed{\begin{aligned} & \frac{m_2 R^2}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \ddot{\varphi} + \frac{m_2 R^2}{8} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ & + \frac{k R^2}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \\ & + \frac{\alpha R^2}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \dot{\varphi} = 0. \end{aligned}} \quad (30)$$

In order to reduce the number of independent parameters and reveal the characteristic scales governing the dynamics, the equations of motion are now written in dimensionless form. The natural length scale of the system is the radius R of the circular track, which provides the characteristic geometric length of the problem. The dimensionless time variable is introduced as

$$\tau = \omega_0 t, \quad \text{with} \quad \omega_0 = \sqrt{\frac{k}{m_1}}. \quad (31)$$

The quantity ω_0 corresponds to the natural frequency of a mass-spring oscillator with mass m_1 and stiffness k , which sets the characteristic time scale of the system. With this choice, the time derivatives transform according to

$$(\dot{\cdot}) = \omega_0 (\cdot)', \quad (\ddot{\cdot}) = \omega_0^2 (\cdot)'',$$

where the prime notation $(\cdot)'$ denotes differentiation with respect to the dimensionless time τ .

Two dimensionless parameters naturally emerge from this process. The first is the mass ratio

$$\mu = \frac{m_2}{m_1}, \quad (32)$$

which characterizes the relative inertia of the two particles. When $\mu \ll 1$, the second mass responds almost instantaneously to the spring force and the dynamics are dominated by the motion of m_1 . Conversely, when $\mu \gg 1$, the elliptical mass is much heavier and the circular mass acts as a light oscillator driven by the motion of m_2 .

The second parameter is the dimensionless friction coefficient

$$\beta = \frac{\alpha}{m_1 \omega_0} = \frac{\alpha}{\sqrt{k m_1}}, \quad (33)$$

which measures the strength of the viscous dissipation acting on m_2 relative to the elastic restoring force. The limit $\beta = 0$ corresponds to the frictionless system, while large values of β indicate that the motion of m_2 is strongly damped.

With this choice of reference quantities, the common factor $m_1 R^2 \omega_0^2 = k R^2$ can be factored out of every term in equations (29) and (30), leaving the system expressed solely in terms of μ , β and the dimensionless time τ .

The dimensionless equation of motion for θ reads

$$\boxed{\theta'' + \frac{1}{8} \frac{l-R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) = 0.} \quad (34)$$

Proceeding similarly with equation (30), the dimensionless equation of motion for φ reads

$$\boxed{\begin{aligned} & \frac{\mu}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi'' + \frac{\mu}{8} \sin \varphi \cos \varphi (\varphi')^2 \\ & + \frac{1}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \\ & + \frac{\beta}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi' = 0. \end{aligned}} \quad (35)$$

In both equations, the ratio $(l-R)/l$ involves the dimensionless spring length l/R , which is directly obtained from equation (28) as

$$\frac{l}{R} = \sqrt{1 - \frac{3}{4} \cos \theta \cos \varphi - \frac{1}{4} \sin \theta \sin \varphi + \frac{1}{64} (9 \cos^2 \varphi + \sin^2 \varphi)}, \quad (36)$$

so that the dimensionless spring extension can be evaluated as

$$\frac{l - R}{l} = 1 - \frac{1}{l/R}. \quad (37)$$

The non-dimensionalization reveals that the dynamics of the system are entirely governed by only two parameters: the mass ratio μ and the damping coefficient β .